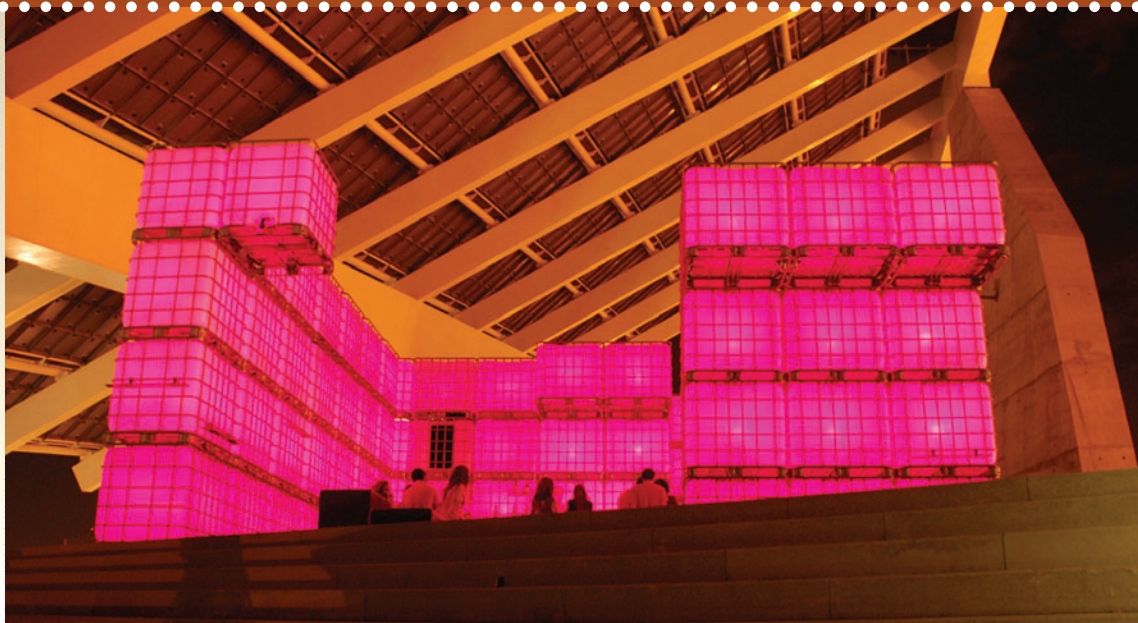


3.2 Perfect Squares, Perfect Cubes, and Their Roots

LESSON FOCUS

Identify perfect squares and perfect cubes, then determine square roots and cube roots.

Kubiks were temporary structures built in Barcelona, Berlin, and Lisbon in 2007.



Make Connections

The edge length of the Rubik's cube is 3 units.

What is the area of one face of the cube? Why is this number a *perfect square*?

What is the volume of the cube? This number is called a **perfect cube**. Why do you think it has this name?



Competitors solve a Rubik's cube in the world championships in Budapest, Hungary, in October 2007.

Construct Understanding

TRY THIS

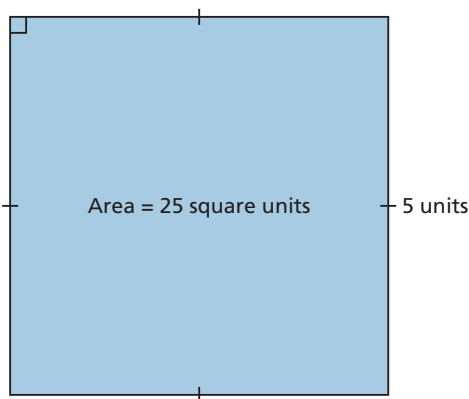
Work in a group.

You will need: 100 congruent square tiles, 100 linking cubes, grid paper, and isometric dot paper.

- A. Use the tiles to model all the perfect squares from 1 to 100. Sketch each model on grid paper and record the corresponding perfect square.
- B. Use the cubes to model all the perfect cubes from 1 to 100. Sketch each model on isometric dot paper and record the corresponding perfect cube.
- C. Which numbers from 1 to 100 are perfect squares and which are perfect cubes?
- D. Join with another group of students. Take turns to choose a number between 101 and 200. Determine if it is a perfect square or a perfect cube.
- E. Suppose you did not have tiles or cubes available. How could you determine if a given number is a perfect square? How could you determine if a given number is a perfect cube?

Any whole number that can be represented as the area of a square with a whole number side length is a *perfect square*.

The side length of the square is the *square root* of the area of the square.



We write: $\sqrt{25} = 5$

25 is a perfect square and 5 is its square root.

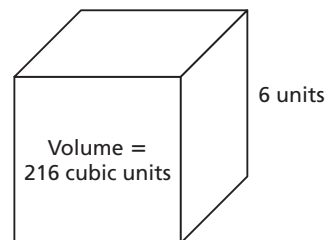
The *square root* of a number n , denoted $\sqrt{n}$, is a positive number whose square is n .

The **cube root** of a number n , denoted $\sqrt[3]{n}$, is a number whose cube is n .

Any whole number that can be represented as the volume of a cube with a whole number edge length is a **perfect cube**. The edge length of the cube is the **cube root** of the volume of the cube.

We write: $\sqrt[3]{216} = 6$

216 is a perfect cube and 6 is its cube root.



Example 1 Determining the Square Root of a Whole Number

Determine the square root of 1296.

SOLUTIONS

Method 1

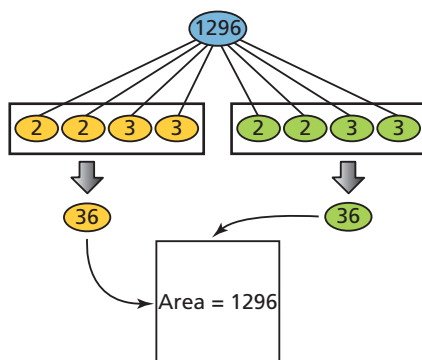
Write 1296 as a product of its prime factors.

$$\begin{aligned} 1296 &= 2 \cdot 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 3 \cdot 3 \\ &= (2 \cdot 2)(2 \cdot 2)(3 \cdot 3)(3 \cdot 3) \\ &= (2 \cdot 2 \cdot 3 \cdot 3)(2 \cdot 2 \cdot 3 \cdot 3) \\ &= 36 \cdot 36 \end{aligned}$$

Group the factors in pairs.
Rearrange the factors in 2 equal groups.

Since 1296 is the product of two equal factors, it can be represented as the area of a square.

So, the square root of 1296 is 36.



Method 2

Estimate:

$$30^2 = 900 \text{ and } 40^2 = 1600$$

$$900 < 1296 < 1600$$

$$\text{So, } 30 < \sqrt{1296} < 40$$

1296 is about halfway between 900 and 1600.

So, $\sqrt{1296}$ is about halfway between 30 and 40.

Use guess and test to refine the estimate.

Try 35: $35^2 = 1225$ (too small, but close)

Try 36: $36^2 = 1296$

So, the square root of 1296 is 36.

CHECK YOUR UNDERSTANDING

- Determine the square root of 1764.

[Answer: 42]

What if the prime factors cannot be grouped into pairs? What can you say about the number?

Example 2

Determining the Cube Root of a Whole Number

Determine the cube root of 1728.

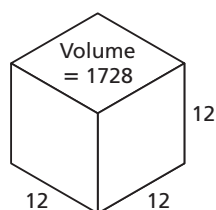
SOLUTIONS

Method 1

Write 1728 as a product of its prime factors.

$$\begin{aligned}1728 &= 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 \cdot 3 \cdot 3 \cdot 3 && \text{Group the factors in sets of 3.} \\&= (2 \cdot 2 \cdot 2)(2 \cdot 2 \cdot 2)(3 \cdot 3 \cdot 3) && \text{Rearrange the factors in} \\&= (2 \cdot 2 \cdot 3)(2 \cdot 2 \cdot 3)(2 \cdot 2 \cdot 3) && \text{3 equal groups.} \\&= 12 \cdot 12 \cdot 12\end{aligned}$$

Since 1728 is the product of three equal factors, it can be represented as the volume of a cube.



So, the cube root of 1728 is 12.

Method 2

Estimate:

$$10^3 = 1000 \text{ and } 20^3 = 8000$$

$$1000 < 1728 < 8000$$

$$\text{So, } 10 < \sqrt[3]{1728} < 20$$

1728 is closer to 1000 than to 8000.

So, $\sqrt[3]{1728}$ is closer to 10 than to 20.

Use guess and test to refine the estimate.

$$\text{Try 11: } 11^3 = 1331 \text{ (too small, but close)}$$

$$\text{Try 12: } 12^3 = 1728$$

So, the cube root of 1728 is 12.

CHECK YOUR UNDERSTANDING

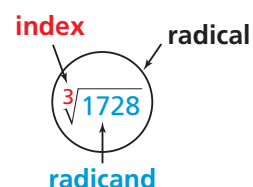
2. Determine the cube root of 2744.

[Answer: 14]

How could you use this strategy to show a number is not a perfect cube?

Can a whole number be both a perfect square and a perfect cube? Explain.

In *Example 1*, $\sqrt{1296}$ is a radical with radicand 1296. The square root can be written as $\sqrt[2]{1296}$, with index 2, but the index on a square root is implied and often not written.



Example 3 Using Roots to Solve a Problem

A cube has volume 4913 cubic inches. What is the surface area of the cube?

SOLUTION

Sketch a diagram.

To calculate the surface area, first determine the edge length of the cube.

The edge length e , of a cube is equal to the cube root of its volume.

$$e = \sqrt[3]{4913}$$

$$e = 17$$

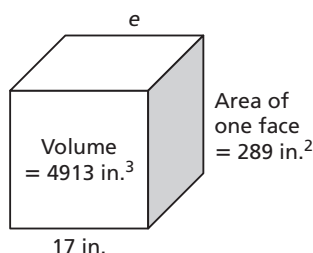
The surface area, SA , of a cube is the sum of the areas of its 6 congruent square faces.

$$SA = 6(17 \cdot 17)$$

$$SA = 6(289)$$

$$SA = 1734$$

The surface area of the cube is 1734 square inches.



CHECK YOUR UNDERSTANDING

3. A cube has volume 12 167 cubic feet. What is the surface area of the cube?

[Answer: 3174 square feet]

Discuss the Ideas

1. What strategies might you use to determine if a number is a perfect square or a perfect cube?
2. What strategy could you use to determine that a number is not a perfect square? Not a perfect cube?
3. What strategies can you use to determine the square root of a perfect square?
What strategies can you use to determine the cube root of a perfect cube?

Exercises

A

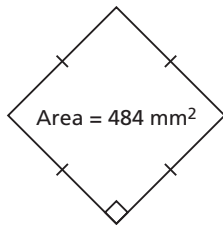
4. Determine the square root of each number.
Explain the process used.
a) 196 b) 256 c) 361 d) 289 e) 441
5. Determine the cube root of each number.
Explain the process used.
a) 343 b) 512 c) 1000 d) 1331 e) 3375

B

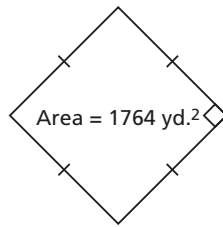
6. Use factoring to determine whether each number is a perfect square, a perfect cube, or neither.
a) 225 b) 729 c) 1944
d) 1444 e) 4096 f) 13 824

7. Determine the side length of each square.

a)

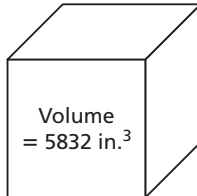


b)

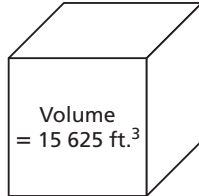


8. Determine the edge length of each cube.

a)



b)



9. In February 2003, the Battlefords Chamber of Commerce in Saskatchewan placed a cage containing a 64-cubic foot ice cube along Yellowhead Highway. Local customers were asked to predict when the ice cube would melt enough for a ball above the ice cube to fall through it. What was the surface area of the cube?



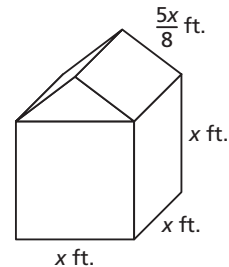
10. A cube has surface area 6534 square feet. What is its volume?
11. Is it possible to construct a cube with 2000 interlocking cubes? Justify your answer.
12. Determine all the perfect square whole numbers and perfect cube whole numbers between each pair of numbers:
- a) 315 – 390 b) 650 – 750
c) 800 – 925 d) 1200 – 1350
13. Write 3 numbers that are both perfect squares and perfect cubes.

14. During the Festival du Voyageur in Winnipeg, Manitoba, teams compete in a snow sculpture competition. Each team begins with a 1440-cubic foot rectangular prism of snow. The prism has a square cross-section and height 10 ft. What are its length and width?



C

15. a) Write an expression for the surface area of this tent. Do not include the floor.



- b) Suppose the surface area of the tent is 90 square feet. Calculate the value of x .
16. Determine the dimensions of a cube for which its surface area is numerically the same as its volume.
17. a) Determine the side length of a square with area $121x^4y^2$.
b) Determine the edge length of a cube with volume $64x^6y^3$.
18. Which pairs of perfect cubes have a sum of 1729?

Reflect

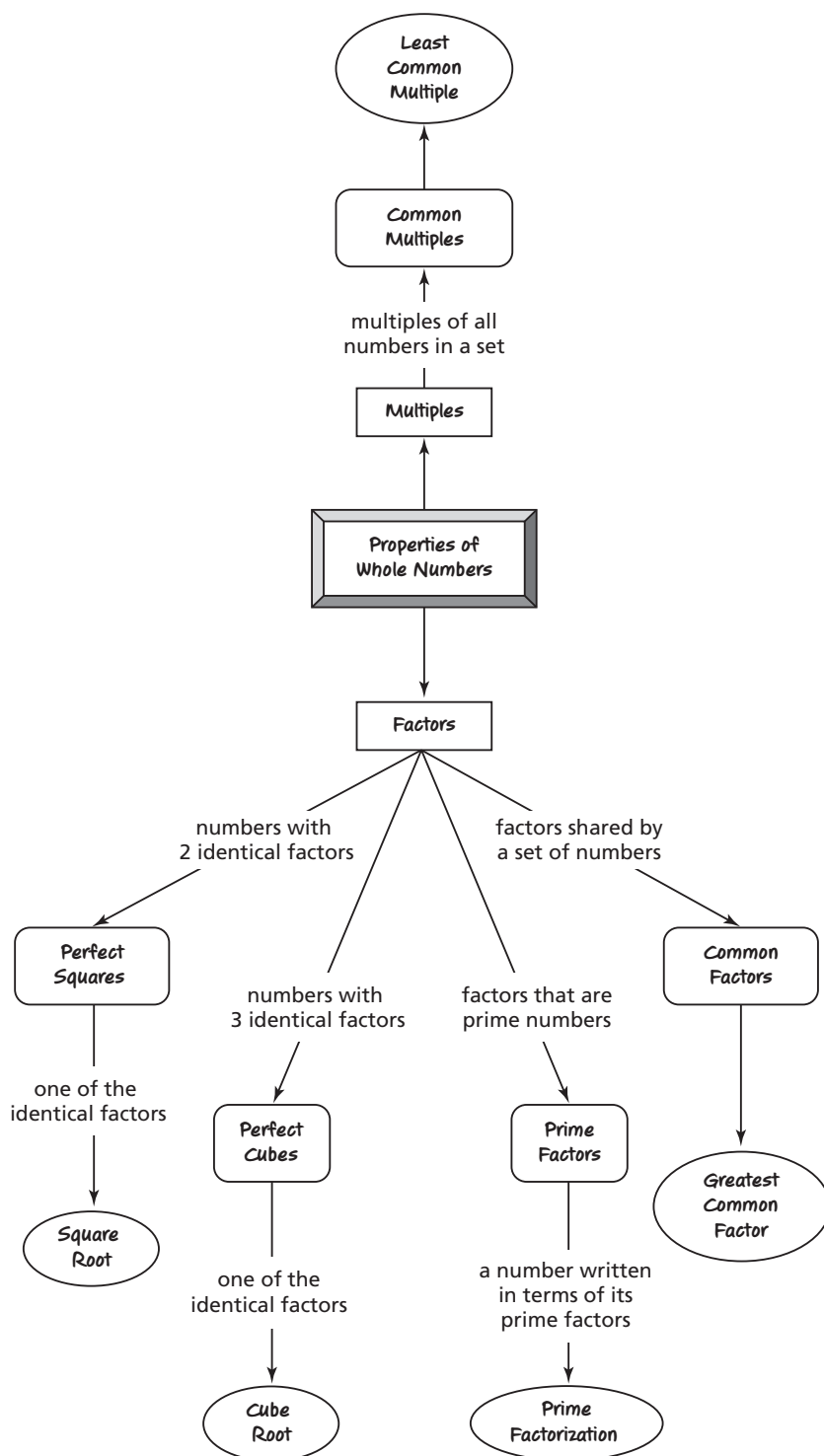
How is determining the square root of a number similar to determining its cube root? How are the strategies different?



CHECKPOINT 1

Connections

Concept Development



■ In Lesson 3.1

- You applied whole number properties and operations to determine **prime factors**.
- You used prime factors to determine **greatest common factor (GCF)** and **least common multiple (LCM)**.

■ In Lesson 3.2

- You used factors and multiples to determine **perfect square whole numbers** and their **square roots**.
- You used factors and multiples to determine **perfect cube whole numbers** and their **cube roots**.

Assess Your Understanding

3.1

- Use powers to write each number as a product of its prime factors.
a) 1260 b) 4224 c) 6120
d) 1045 e) 3024 f) 3675
- Determine the greatest common factor of each set of numbers.
a) 40, 48, 56 b) 84, 120, 144
c) 145, 205, 320 d) 208, 368, 528
e) 856, 1200, 1368 f) 950, 1225, 1550
- Determine the least common multiple of each set of numbers.
a) 12, 15, 21 b) 12, 20, 32 c) 18, 24, 30
d) 30, 32, 40 e) 49, 56, 64 f) 50, 55, 66
- Use the least common multiple to help determine each answer.
a) $\frac{8}{3} + \frac{5}{11}$ b) $\frac{13}{5} - \frac{4}{7}$ c) $\frac{9}{10} \div \frac{7}{3}$
- The Mayan used several different calendar systems; one system used 365 days, another system used 260 days. Suppose the first day of both calendars occurred on the same day. After how many days would they again occur on the same day? About how long is this in years? Assume 1 year has 365 days.

3.2

- Determine the square root of each number. Which different strategies could you use?
a) 400 b) 784 c) 576
d) 1089 e) 1521 f) 3025
- Determine the cube root of each number. Which different strategies could you use?
a) 1728 b) 3375 c) 8000
d) 5832 e) 10 648 f) 9261
- Determine whether each number is a perfect square, a perfect cube, or neither.
a) 2808 b) 3136 c) 4096
d) 4624 e) 5832 f) 9270
- Between each pair of numbers, identify all the perfect squares and perfect cubes that are whole numbers.
a) 400 – 500 b) 900 – 1000 c) 1100 – 1175
- A cube has a volume of 2197 m^3 . Its surface is to be painted. Each can of paint covers about 40 m^2 . How many cans of paint are needed? Justify your answer.